INCH-POUND

MIL-PRF-49142/108
12 January 1990
SUPERSEDING
MIL-C-49142/10A
21 April 1988

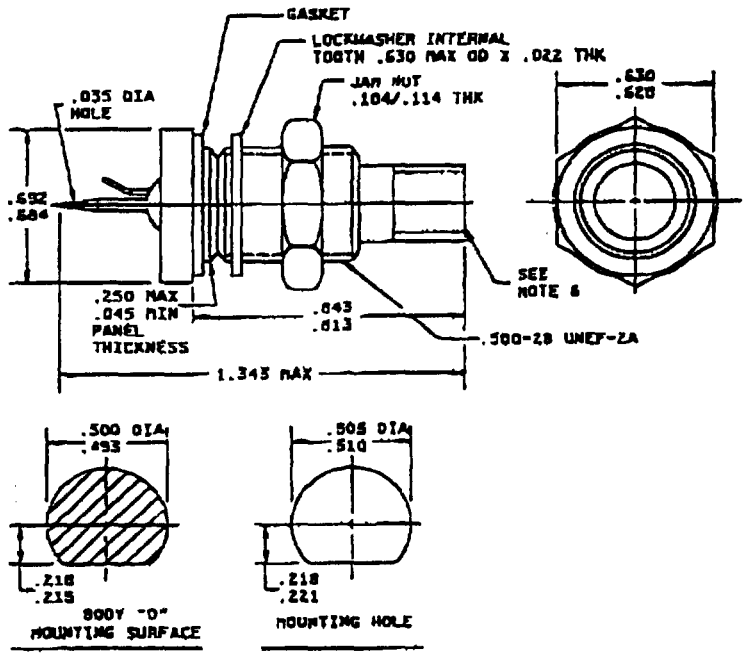
NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

PERFORMANCE SPECIFICATION

CONNECTORS, TRIAXIAL, RADIO FREQUENCY, (SERIES TRT (CABLED) RECEPTACLE, SOCKET CONTACT, JAMNUT MOUNTED, CLASS 2)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-PRF-49142



Inches	mm
.022	.56
.035	.88
.045	1.14
.104	2.64
.114	2.90
.215	5.46
.218	5.54
.221	5.61
.250	6.35
.493	12.52
.500	12.70
.505	12.71
.510	12.95
.620	15.75
.630	16.00
.684	17.37
.692	17.58
.813	20.65
.843	21.41
1.343	34.11

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Wrench flats are to accommodate standard wrench openings per FED-STD-H28, appendix 10.
4. All undimensioned pictorial representations are for reference purposes only.
5. Unless otherwise specified all tolerances are ±.005 inch.
6. Interface per MIL-STD-348, series TRT, socket contact.

FIGURE 1. General configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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ENGINEERING PARAMETERS.

Nominal impedance Nonconstant

Frequency range 0 to 500 MHz minimum

Voltage rating: 400 V rms, maximum working voltage at sea level. 100 V rms, maximum working voltage at 70,000 feet.

Temperature range. -65°C to +165°C.

REQUIREMENTS:

Dimensions and configuration. See figure 1 and MIL-STD-348.

Force to engage and disengage:
Longitudinal force: Not applicable.
Torque 2.5 inch-pounds maximum.

Coupling proof torque: Not applicable.

Inspection conditions: Coupling torque: 4-6 inch-pounds.

Mating characteristics. See figure 1 and MIL-STD-348 for dimensions.
Center contact (socket):
Oversize test pin: 0.040 inch diameter, minimum (nonclosed entry contacts only)
Insertion depth: .125 inch minimum.
Number of insertions: 1.
Insertion force test:
Steel test pin finish: 16 microinches.
Insertion force 2 pounds maximum.
Steel test pin diameter. .039 minimum.
Withdrawal force test:
Steel test pin diameter. .037 maximum.
Withdrawal force 2 ounces minimum
Test pin finish 16 microinches.

Permeability: Applicable.

Hermetic seal: See table I.

Leakage Not applicable.

Insulation resistance: 5000 megohms.

Conductor retention 6 pounds minimum, axial force.

Dielectric withstanding voltage At sea level, 1200 V rms, between center conductor and intermediate conductor. 500 V rms, between intermediate conductor and outer conductor.

Salt spray (corrosion): Applicable.

Vibration. Applicable

Shock Applicable

Thermal shock Applicable.

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Moisture resistance Applicable

Conductor resistance In milliohms, maximum

	Connector dash number	Initial	After environment
Center conductor	0001	2.0	2 5
	0002	12.0	12 5
Intermediate conductor	0001	0.5	0.6
	0002	1.0	1.5
Outer conductor	0001	0.2	0.3
	0002	0.5	0.6

Corona level
Altitude. 70,000 feet.
Voltage: 200 V rms minimum

RF high potential withstanding voltage.
800 V rms, between center conductor and intermediate conductor.
200 V rms, between intermediate conductor and outer conductor at 5 to 7.5 MHz.

Leakage current: Not applicable.

Cable retention force (for cable .200 to .325 outside dimension) 40 pounds, minimum

Coupling mechanism retention force Not applicable.

Risetime degradation Not applicable

Connector durability· 500 cycles minimum at 12 cycles per minute maximum.

Part or Identifying Number (PIN). M49142/10- (dash number from table I.)

TABLE I. Dash number, superseded PIN.

Dash number	Superseded PIN
-0001	---
-0002 (hermetic)	---

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians
Army - CR
Navy - EC
Air Force - 85

Review activities.
Army - AR, MI
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AT
Navy - AS, MC
Air Force - 19

Preparing activity
Army - CR

Agent
DLA - ES

(Project 5935-3750-02)

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